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PAPER

Elastic instabilities in a microfluidic cross-slot flow of wormlike micellar solutions†

Neville Dubash, Perry Cheung and Amy Q. Shen*

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We present a series of experiments involving an elastic instability that occurs with the flow of wormlike micellar solutions in a microfluidic cross-slot device. We use four different concentrations of an aqueous solution of cetyltrimethylammonium bromide (CTAB) and sodium salicylate (NaSal); two highly viscoelastic and two weakly viscoelastic. Flow in the microfluidic cross-slot device is examined using birefringence and PIV measurements. With all of the solutions we observe the formation of a birefringent band along the outflow axis, the intensity of which increases with the flow rate. In the two highly viscoelastic solutions, as the flow rate increases, the instability results in flow transitioning from a stable symmetric flow to a stable asymmetric flow, to an unsteady asymmetric flow. With the weakly viscoelastic solutions the instability results in the flow transitioning directly from a stable symmetric flow to an unsteady flow. The critical Weissenberg numbers at which the transitions occur, increase with increasing elasticity number (defined as the ratio of the Weissenberg number to the Reynolds number). With the two highly viscoelastic solutions we also observe the formation of lip vortices along the walls of the inlet channels. For one of the highly viscoelastic solutions the symmetry-breaking occurs before the formation of the secondary flow, whereas in another solution the secondary flow develops prior to the symmetry-breaking. This indicates that these instabilities are complex phenomena, whose behavior is likely influenced by a combination of factors such as Weissenberg number, inertial effects, and rheological behavior of the micellar solutions.

1 Introduction

In this work we present details of an elastic instability that occurs in a microfluidic cross-slot flow of wormlike micellar solutions. The instability results in a symmetry-breaking of the velocity field, where there can be either a steady asymmetric flow or an unsteady oscillatory flow pattern. Additionally, under certain conditions a secondary flow develops, which results in the appearance of vortices in the inlet channels.

Micelles have the ability to break and reform dynamically during flow and even when quiescent. Micelles that break due to strain from flow can reform once flow ceases; it is for this reason that micellar solutions are often referred to as "living" polymers. Moreover, certain flows can cause micellar alignment, which in turn can cause the micelles to grow and form oriented structures. Micellar solutions can exhibit temporary structure formation,[1–4] shear thickening[5–7] and shear banding.[8–11]

The early work by Gardner *et al.*[12] used cross-slot geometries to investigate the effects of polymer microstructure and fluid–wall interaction on the flow behavior of polystyrene by using photon-correlation velocimetry and birefringence. Elastic instabilities in a microfluidic cross-slot, similar to those observed here, were first observed with viscoelastic fluids by Arratia *et al.*[13] and Pathak and Hudson.[14] Arratia *et al.* showed that above a critical strain rate molecular stretching in polyacrylamide solutions can result in two instabilities, the first where the velocity field becomes asymmetric, and the second where the velocity field fluctuates nonperiodically in time. Pathak and Hudson, while examining the birefringent behavior of micellar solutions, also observed that at high flow rates the birefringent band that forms along the outflow axis becomes angled relative to the outflow axis, and that the velocity field becomes asymmetric. These phenomena were observed for Reynolds numbers smaller than 10^{-2}, and thus could not be inertially driven. Most recently, Haward *et al.*[15] performed some detailed experiments using a micellar solution (cetylpyridinium chloride and sodium salicylate) in a microfluidic cross-slot device. They observed that the elastic instabilities appeared to be more pronounced than those occurring in polymer solutions, most likely because of the shear localization which can occur with micellar solutions.

Numerical simulations of viscoelastic fluids in a microfluidic cross-slot have also shown similar symmetry-breaking instabilities. These simulations have been done at both zero and low

Mechanical Engineering Department, University of Washington, Seattle, WA, 98195, USA. E-mail: amyshen@uw.edu; Fax: +206 685 8047; Tel: +206 221 0372

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Reynolds number with a variety of viscoelastic constitutive models, including the upper-convected Maxwell model and finite extendible nonlinear elastic (FENE) models.[16–18] It was shown that the instability can occur in the absence of inertia, and that inertial effects tended to suppress the instabilities. However, in the numerical simulations, the development of the instabilities (as the flow rate increases) has not always been observed to be the same. In some cases, the steady asymmetric flow regime was not observed, and flow transitioned directly from steady symmetric flow to unsteady flow. It has also been shown that the extensional properties (extensibility parameter) and the solvent viscosity ratio used in the numerical models can affect the critical threshold for transition.[17]

A similar transition from steady symmetric flow to steady asymmetric flow to unsteady flow was observed by Soulages *et al.*[19] for a polyethylene oxide solution in a T-shaped microchannel that contained a recirculating cavity region. For a T-shaped microchannel that lacked the recirculating cavity the steady asymmetric flow regime was absent and flow transitioned from steady symmetric to unsteady. They concluded that in the free stagnation point flow (with the recirculating cavity) a highly oriented phase is formed in the region of extensional flow, and that it is this structured region that leads to the symmetry-breaking. In the case when the stagnation point is pinned (no recirculating cavity) the symmetry-breaking instability was suppressed and the system would transition directly from steady symmetric flow to unsteady flow. Also, in some related work, involving somewhat different geometries or configurations than that used here, researchers have started to map out the behavior of these viscoelastic instabilities over a portion of the parameter space. In these cases, it was found that factors such as the Reynolds number, relative channel size, and relative channel flow rates all affect whether the steady asymmetric flow regime is observed.[20,21]

Separate from the aforementioned symmetry-breaking phenomenon, which appears to occur with several different types of viscoelastic solutions or models, a micelle-rupture phenomenon has also been observed in experiments.[22,23] For certain micellar solutions, when they are exposed to an extensional rate above a critical value, larger than that which the micelles can withstand, a catastrophic breaking of the micellar chains occur. It has been shown that this micelle-rupture can also lead to instabilities in certain flows.[23,24]

Finally, secondary flows which result in the formation of lip vortices have been observed in microfluidic flows of viscoelastic fluids. In microfluidic contraction–expansion geometries, elasticity results in the formation of a variety of vortices in the upstream channel.[25,26] And, more closely related to the cross-slot geometry, viscoelastic flows around 90° bends have been observed to develop a secondary flow which results in a lip vortex in the upstream channel.[27,28]

In this work, we study the symmetry-breaking instability in a microfluidic cross-slot device using four different concentrations of a micellar solution. Two of the concentrations are strongly viscoelastic, and exhibit strong shear-thinning. The other two concentrations are weakly viscoelastic and exhibit some shear-thickening at higher shear rates, followed by mild shear-thinning. In section 2 we provide a description of the micellar solutions and of the experimental setup. The results of the experiments are detailed in section 3, and a discussion of the results is presented in section 4. We close in section 5 with conclusions.

2 Experimental description

2.1 Micellar solutions

The wormlike micellar solutions used in this work were aqueous solutions of cetyltrimethylammonium bromide (CTAB) and sodium salicylate (NaSal). The solutions were prepared by adding the appropriate amount of CTAB and NaSal to deionized water and mixing for a minimum of 24 h, and then leaving the solutions to sit overnight to equilibrate. Four different solutions of varying CTAB and NaSal concentration were examined in this work: CTAB concentrations of 0.050 M, 0.075 M, and 0.100 M with NaSal to CTAB molar ratios of $R = 0.32$ were used, and also a solution with a CTAB concentration of 0.075 M with $R = 0.24$.

The rheology of the 0.100 M CTAB and 0.075 M CTAB, $R = 0.32$ solutions was measured using a stress-controlled rotational rheometer (TA Instruments AR2000) with a 60 mm degree^{-1} cone and plate geometry. The 0.050 M CTAB, $R = 0.32$ and the 0.075 M, $R = 0.24$ solutions were very weakly viscoelastic, and as a result the rheometer was not sensitive enough to measure the linear viscoelastic behavior. For these two solutions the linear viscoelastic behavior was instead measured by using microrheology.[29] In general, CTAB–NaSal solutions are viscoelastic fluids and exhibit a range of interesting phenomenon under certain conditions, some of which are visible in Fig. 1. The 0.100 M CTAB and the 0.075 M CTAB, $R = 0.32$ solutions exhibit strong shear thinning while the 0.050 M, $R = 0.32$ and the 0.075 M CTAB, $R = 0.24$ solutions exhibit shear thickening at high shear rates followed by slightly shear thinning, as shown in Fig. 1a. The shear stress *versus* shear rate curves (Fig. 1b) show a stress plateau for the 0.100 M CTAB and the 0.075 M CTAB, $R = 0.32$ solution which is indicative of shear banding,[30–32] while the 0.050 M, $R = 0.32$ and 0.075 M CTAB, $R = 0.24$ solutions do not appear to show any indication of a stress plateau or shear banding.

Fig. 2a shows the linear viscoelastic behavior for the 0.100 M CTAB and 0.075 M CTAB, $R = 0.32$ solution and the corresponding fits to a single-mode Maxwell model *via* the G' (storage moduli) and G'' (loss moduli) curves *versus* oscillation frequency ω. The linear visocelastic behavior of the 0.050 M, $R = 0.32$ and the 0.075 M CTAB, $R = 0.24$ solutions did not match the single-mode Maxwell model very well and are not shown. Instead, normalized Cole–Cole plots for all the solutions are shown in Fig. 2b with the solid black curve corresponding to a pure Maxwellian fluid. Based on Fig. 2b, we can see that even the most viscoelastic 0.100 M CTAB solution does not behave very Maxwellian and the less viscoelastic solutions deviate even further. This deviation can be attributed to a spectrum of relaxation times associated with various wormlike micelles of varying length and stability due to the low salt to surfactant ratio.[33]

Table 1 lists the longest relaxation time, λ, the plateau modulus, G_0, and the zero-shear viscosity, η_0, of the CTAB–NaSal solutions used here. The relaxation time was determined

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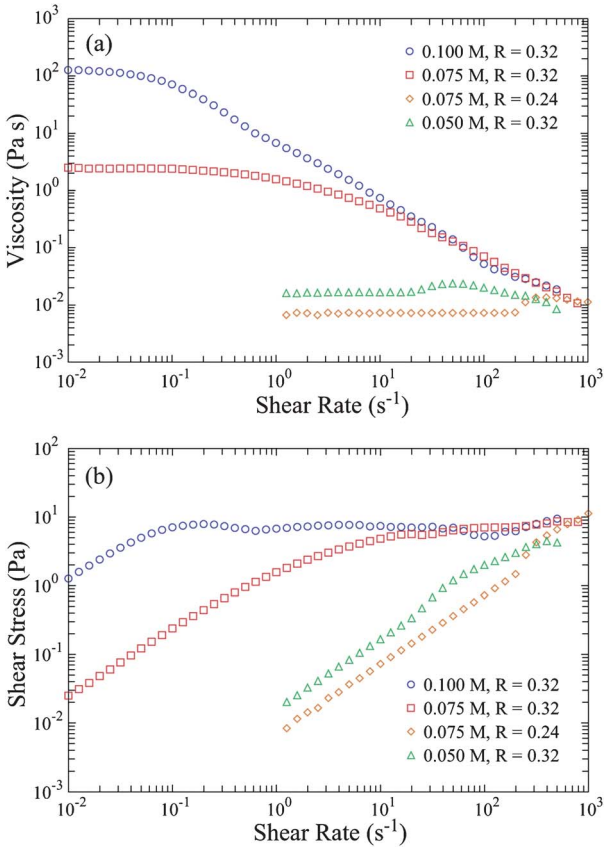

Fig. 1 (a) Shear viscosity and (b) shear stress as a function of the shear rate for the micellar solutions under steady shear.

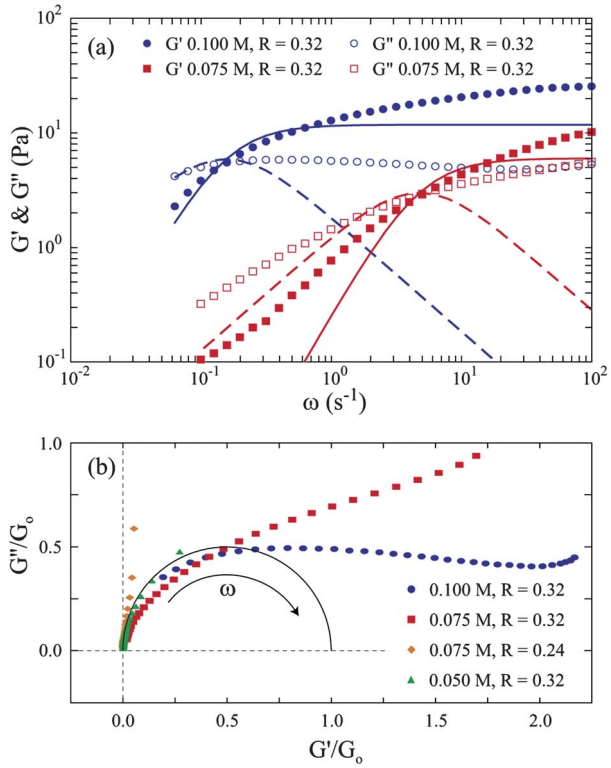

Fig. 2 (a) The storage modulus (G', filled symbols) and loss modulus (G'', open symbols) as a function of frequency for the 0.100 M CTAB and the 0.075 M CTAB, $R = 0.32$ solutions under oscillatory shear in the cone and plate geometry. The data has been fitted with a single-mode Maxwell model (lines). (b) Normalized Cole–Cole plot using the data from (a) and also from the microrheology data for the 0.050 M, $R = 0.32$ and 0.075 M CTAB, $R = 0.24$ solutions.

from the raw data of the first cross-over region of G' and G'' curves, while the plateau modulus G_0 was subsequently determined from fitting the G' and G'' curves to a single-mode Maxwell model, and the zero shear viscosity was determined from the steady-state shear viscosity measurements. Local relaxation times based on the normal stress difference for the 0.100 M and the 0.075 M CTAB, $R = 0.32$ solutions can be calculated by $\lambda(\dot{\gamma}) = \frac{N_1(\dot{\gamma})}{2\tau_{xy}(\dot{\gamma})}$ and then taking the averaged relaxation values at low shear rates. At the limit of low shear rates, the local relaxation times for the 0.100 M CTAB and 0.075 M CTAB, $R = 0.32$ solutions are 2.1 s and 0.37 s respectively, which are comparable to the relaxation time from our Maxwell fit of 6.4 s and 0.21 s respectively, as indicated in Table 1. Nonlinear viscoelastic behavior of the 0.100 M CTAB and 0.075 M CTAB, $R = 0.32$ solutions using large amplitude oscillatory shear (LAOS) measurements is observed to be largely strain thinning, details are supplied as supplementary material.

Table 1 The longest relaxation time, λ, plateau modulus G_0, and the zero-shear viscosity η_0 for the micellar solutions

CTAB conc. (M)	R	λ (s)	G_0	η_0 (Pas)
0.050	0.32	0.008	2.0	0.016
0.075	0.24	0.008	1.7	0.008
0.075	0.32	0.21	6.0	2.5
0.100	0.32	6.4	12.0	126

2.2 Experimental setup

The microfluidic cross-slot device was fabricated from acrylic. The channels were machined onto one side of a 1/4″ thick acrylic block (Tap Plastics, Seattle, WA) and four inlet holes at the end of each arm of the cross were drilled and tapped. The machined block was then bonded to a 1/16″ thick base sheet of acrylic using acrylic cement (Tap Plastics). The threaded input ports allowed us the flexibility to use threaded fittings (Upchurch Scientific) which are capable of high pressures and permit easy swapping of fluidic components. The channel width was 500 μm and the depth was 250 μm; each arm was 30 mm long, which was sufficiently long to allow the flow to become fully developed after entering through the inlet port (see Fig. 3). A syringe pump (Harvard Apparatus) was used to pump the solution from a single syringe to the two inlet ports *via* a standard "Y" connector (Upchurch Scientific). Note that in this work, when we refer to the flow rate, this is the value of the flow rate in a single inlet channel; the net flow rate into the microfluidic cross-slot device is twice this quantity.

The microfluidic device was imaged on an inverted microscope (Leica DMIRB and Leica DMI3000) and images were taken with a high-speed monochrome camera (Vision Research Phantom

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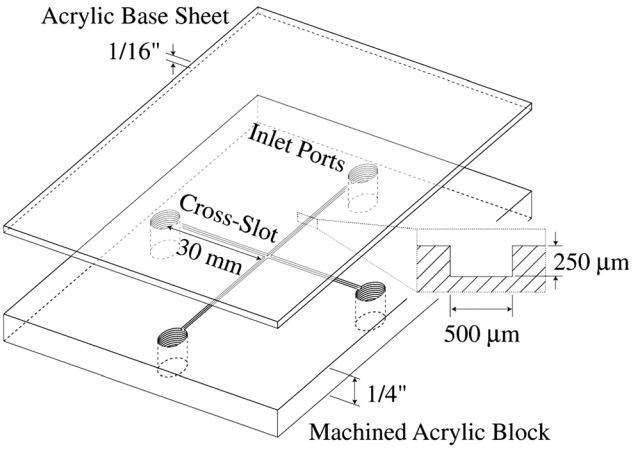

Fig. 3 A schematic of the microfluidic cross-slot device.

v310). Images were taken with a 10× NA 0.22 objective with the focal plane at the middle of the channel. The birefringence measurements were taken with cross polarizers oriented at 45 degrees and 135 degrees from the inlet channel axis (which corresponds to the vertical and horizontal directions in all of the birefringence images shown).

For one of the solutions, the 0.100 M, $R = 0.32$ CTAB solution, particle image velocimetry (PIV) was carried out in order to better understand some of the flow characteristics. For the PIV measurements 3 µm diameter polystyrene microspheres (Polysciences) were added to the solution at a concentration of 0.05 wt%. The presence of the microspheres did not noticeably affect the rheology of the solution. Images for PIV were taken in the brightfield with a 10× NA 0.22 objective and had a correlation depth of 61 µm.[34] The steady-state velocity profiles from the PIV analysis were calculated by averaging over 500 images.

3 Results

One of the primary dimensionless parameters when considering flows of viscoelastic fluids is the Weissenberg number,

$$\mathrm{Wi} = \dot{\varepsilon}\lambda, \tag{1}$$

where $\dot{\varepsilon}$ is the rate of extension and λ is the characteristic relaxation time of the viscoelastic fluid. Here, we use the values of λ listed in Table 1, and take $\dot{\varepsilon}$ to be the nominal rate of extension in the cross-slot,

$$\dot{\varepsilon} = \frac{U}{w} = \frac{Q}{w^2 h} \tag{2}$$

where U is the average velocity in a single channel, Q is the flow rate in the channel, w is the width of the channel, and h is the height of the channel. This is generally what is used in the literature (sometimes with an additional factor of 2), and thus allows us to compare the critical Wi values at which various transitions are observed to occur. It has also been shown that the Reynolds number,

$$Re = \frac{\rho U w}{\eta_0}, \tag{3}$$

where ρ is the fluid density and η_0 is the zero-shear viscosity, can affect the flow behavior. Higher Reynolds numbers can push transitions to higher critical Wi values, or can eliminate certain behavior entirely.[16,21] Finally, another useful dimensionless quantity is the elasticity number,

$$El = \frac{Wi}{Re} = \frac{\eta_0 \lambda}{\rho w^2}, \tag{4}$$

which provides a measure of the relative importance of elastic effects to inertial effects. It also has the added benefit that, as defined here, it is constant for a given fluid flowing in a given cross-slot device (independent of the fluid velocity or flow rate). For our two strongly viscoelastic solutions (0.100 M and 0.075 M, $R = 0.32$), $El > 10^3$ and so the elasticity is the dominant effect. For the two weakly viscoelastic solutions (0.050 M, $R = 0.32$ and 0.075 M, $R = 0.24$), $El \sim 1$ and thus elasticity and inertia are relatively balanced.

The first solution that we consider is the 0.100 M CTAB solution; for this solution $El = 3 \times 10^6$. At this concentration the micellar solution exhibits highly viscoelastic behavior and fracturing phenomena can frequently be observed.[22,35] At very low flow rates, $Q < 0.5$ mL h^{-1} ($\dot{\varepsilon} < 2.2$ s^{-1}, $Wi < 14$), the velocity field is steady and symmetric (see Fig. 4). Also, at these low flow rates there is no noticeable birefringence. As the flow rate increases to around $Q = 1$ mL h^{-1} in each inlet channel ($\dot{\varepsilon} = 4.4$ s^{-1}, $Wi = 28$), the flow remains steady and symmetric, and a thin birefringent band begins to form along the outflow axis (Fig. 5a). This birefringent band increases in size and intensity as the flow rate continues to increase. In this paper, we mostly show birefringence images converted to false-color images; this makes small variations in the birefringence much more visible and easily discernible. However, all analysis was done with the original grayscale images.

Eventually at about 2 mL h^{-1} ($\dot{\varepsilon} = 8.9$ s^{-1}, $Wi = 57$), the velocity field begins to become asymmetric, and the asymmetry becomes more pronounced with a further increase of the flow rate (see Fig. 5). At higher flow rates (3 mL h^{-1}, $\dot{\varepsilon} = 13$ s^{-1}, $Wi = 85$) two lip vortices begin forming along the wall of the inlet channels. They appear to form preferentially on the wall closest to the outlet to which the majority of the flow from that inlet is

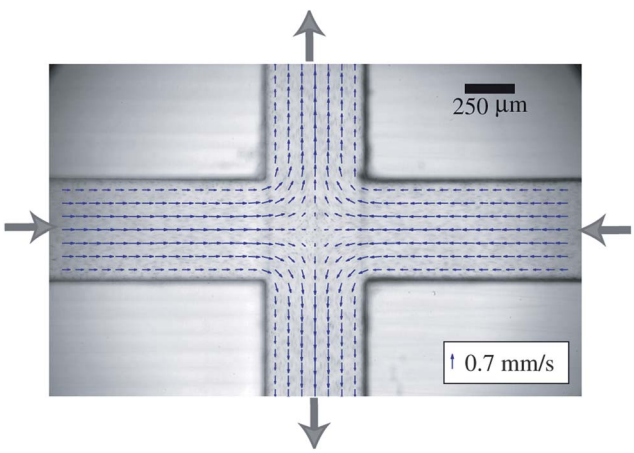

Fig. 4 The velocity field in the cross-slot at a flow rate of 0.2 mL h^{-1} ($\dot{\varepsilon} = 0.9$ s^{-1}, $Wi = 5.7$) with the 0.100 M CTAB, $R = 0.32$ solution.

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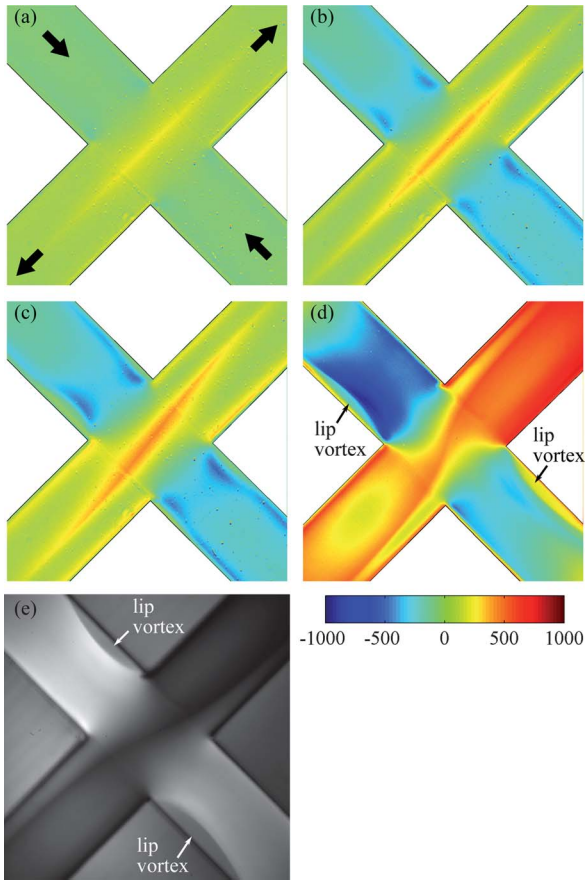

Fig. 5 False-color images of the birefringence of steady symmetric and steady asymmetric flow in the microfluidic cross-slot devices with the 0.100 M CTAB, $R = 0.32$ solution. (a) $Q = 1$ mL h^{-1}, (b) $Q = 2$ mL h^{-1}, (c) $Q = 4$ mL h^{-1}, (d) $Q = 15$ mL h^{-1}, and (e) a true-color grayscale image for $Q = 10$ mL h^{-1}, clearly showing the lip vortices in the inlet channels. Fluid enters from top left and bottom right channels and exits from the top right and bottom left channels. The color scale is arbitrary.

diverted. Examples of steady asymmetric velocity fields with lip vortices are shown in Fig. 6 and 7. In both Fig. 6 and 7 flow from the left inlet channel preferentially enters to the top outlet channel, and flow from the right inlet channel preferentially flows to the bottom outlet channel. That the asymmetry occurs in the same direction is merely a coincidence, when the asymmetry first develops it is equally likely to occur in either direction (*cf.* Fig. 5d,e). Also, the flow along the top wall in the left inlet channel (and bottom wall in the right inlet channel) appears to separate from the wall just prior to bending around the corner towards the outlet channel. Along that section of the wall the velocities are substantially lower, and it is here that the lip vortices form. A close up of one of the lip vortices is provided in Fig. 6b. It can be seen that the velocities here are about two orders of magnitude smaller than those in the bulk flow. The vortices also display birefringence; and while they are somewhat difficult to distinguish in the false-color birefringence images (Fig. 5c,d), they are clearly demarcated in the true grayscale birefringence image along the upper wall of the top left inlet and the lower wall of the bottom right inlet (Fig. 5e, the two darker regions along the inlet walls).

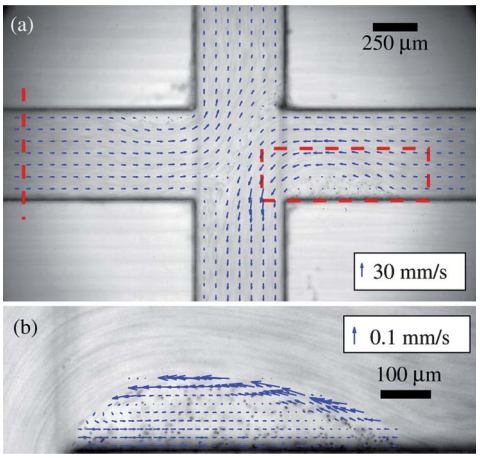

Fig. 6 The velocity field in the cross-slot at a flow rate of 4 mL h^{-1} ($\dot{\varepsilon} = 18$ s^{-1}, $Wi = 114$) with the 0.100 M CTAB, $R = 0.32$ solution. (a) The full cross-slot device, (b) a close up of the velocity field inside the vortex on the lower right of the right inlet channel (the boxed region in (a)). The dashed line in the left inlet channel marks the location at which the velocity profiles in Fig. 9 are measured. The velocities in the vortex are so much smaller than those in the bulk flow that the two cannot be simultaneously resolved using PIV.

At flow rates around 10–30 mL h^{-1} the flow develops a slight "pulsing" unsteadiness, where the orientation of the birefringent band and the size of the vortices begin to fluctuate slowly. Eventually, above $Q = 30$ mL h^{-1} ($\dot{\varepsilon} = 133$ s^{-1}, $Wi = 853$), the flow becomes chaotic and unsteady, with the flow pattern to the outlet channels flipping directions occasionally, and then more frequently as the flow rate continues to increase. In Fig. 8 we show an image of the average birefringence, averaged over 100 images, and the deviation from this average at several instances in time. The deviation images provided a clearer representation of the unsteady behavior. We find that the flow never becomes fully asymmetric (with flow from one inlet going entirely to one outlet), rather the system becomes unsteady first. Of the experiments performed by others, only Haward *et al.*[15] observed a completely asymmetric velocity field.

Fig. 9 shows the normalized velocity profiles across the inlet channel, at a distance of 1240 μm upstream from the stagnation point (this location is marked in Fig. 6). At low flow rates the velocity profile is close to parabolic, as it would be for a Newtonian fluid. As the flow rate increases the profile flattens in the

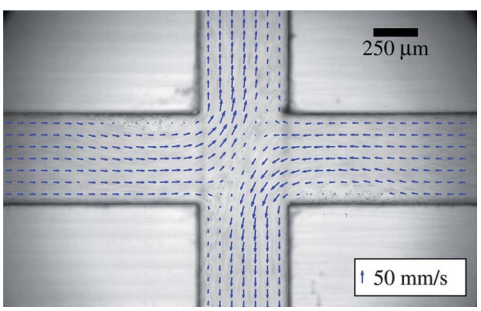

Fig. 7 The velocity field in the cross-slot at a flow rate of 10 mL h^{-1} ($\dot{\varepsilon} = 44$ s^{-1}, $Wi = 284$) with the 0.100 M CTAB, $R = 0.32$ solution.

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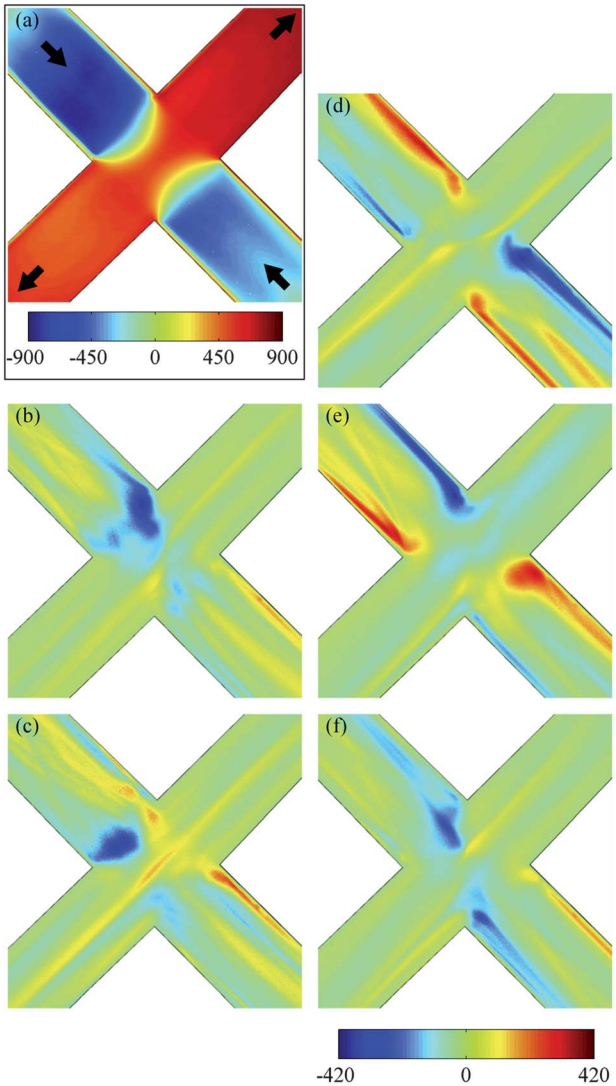

Fig. 8 False-color birefringence images of the unsteady flow which occurs in the microfluidic cross-slot device. The solution is the 0.100 M CTAB, $R = 0.32$ solution, and the flow rate is Q = 70 mL h^{-1}. Image (a) shows the birefringence intensity averaged over 100 images (the frame rate is 100 fps). Images (b)–(f) show the deviation in the birefringence intensity from the average at fixed times: (b) t = 0.0 s, (c) t = 0.1 s, (d) t = 0.2 s, (e) t = 0.3 s, and (f) t = 0.4 s. Fluid enters from the top right and bottom left channels and exits from the top left and bottom right channels.

center of the channel and high-shear layers form near the walls, as is typical of shear-thinning fluids. Additionally, at 10 mL h^{-1} the upstream flow is beginning to become influenced by the asymmetric flow field downstream (here the majority of the flow from this channel is diverted to the left).

For the 0.075 M CTAB, $R = 0.32$ solution, which is also strongly viscoelastic ($El = 2 \times 10^3$), the flow behavior is qualitatively similar, and all the same phenomena are observed. However, the critical Wi value at which each phenomena appears is different. At very low flow rates the velocity field starts out steady and symmetric. Again, at a flow rate of around 1 mL h^{-1} ($\dot{\varepsilon} = 4.4$ s^{-1}, $Wi = 0.9$), a birefringent band appears along the center of the outflow axis. As before, the intensity of this band

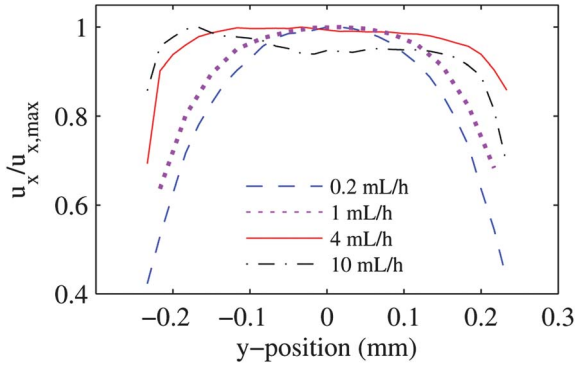

Fig. 9 The normalized velocity profiles, for the 0.100 M CTAB, $R = 0.32$ solution, across the inlet channel at a distance of 1240 μm upstream from the stagnation point (as marked in Fig. 6).

increases with the flow rate (see Fig. 10). However, in contrast to the behavior with the 0.100 M CTAB, $R = 0.32$ solutions, the first phenomena to appear is the formation of the lip vortices at a flow rate of about $Q = 4$ mL h^{-1} ($\dot{\varepsilon} = 18$ s^{-1}, $Wi = 3.7$). In this case, four lip vortices form along the wall of the inlet channels

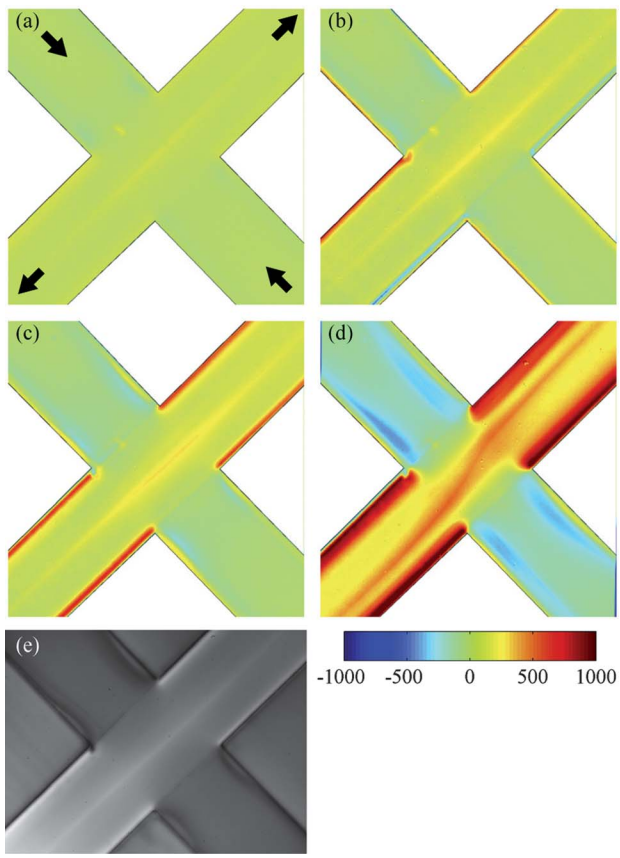

Fig. 10 Birefringence images for the 0.750 M CTAB, $R = 0.32$ solution. False-color images for (a) $Q = 2$ mL h^{-1}, (b) $Q = 6$ mL h^{-1}, (c) $Q = 10$ mL h^{-1}, (d) $Q = 30$ mL h^{-1}, and (e) a true-color grayscale image for $Q = 10$ mL h^{-1}, showing the four lip vortices in the inlet channels. Fluid enters from top left and bottom right channels and exits from the top right and bottom left channels. For (a), (b), (c), and (e) the flows are steady; for (d) the flow is unsteady and the asymmetry flips back and forth chaotically.

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(just visible in Fig. 10b,c,d and more clearly in Fig. 10e). Shortly after the vortices appear, the flow becomes asymmetric (6 mL h^{-1}, $\dot{\varepsilon} = 27\ s^{-1}$, $Wi = 5.6$). With increasing flow rate the asymmetry becomes more pronounced until at around $Q = 20$ mL h^{-1} ($Wi = 19$) the flow becomes unstable, with the direction of asymmetry flipping back and forth chaotically.

With the 0.075 M CTAB, $R = 0.32$ solution the vortices do not extend as far into the channel as with the 0.100 M CTAB solution. As the flow rate increases the vortices appear to flatten out further until they become almost imperceptibly thin. Another difference with the 0.075 M CTAB, $R = 0.32$ solution, is the appearance of birefringent fluid along the walls of the channel. These bands are visible in Fig. 10c,d (*cf.* Fig. 5), and actually begin to become noticeable at a flow rate of $Q = 8$ mL h^{-1} ($\dot{\varepsilon} = 36\ s^{-1}$, $Wi = 7.5$). The bands appear to form upstream in the inlet channels, bend tightly around the corners, and continue into the outlet channels.

The remaining two solutions, the 0.050 M CTAB, $R = 0.32$ and 0.075 M CTAB, $R = 0.24$ solutions, are weakly viscoelastic, with $El = 0.51$ and $El = 0.26$ respectively. With the 0.050 M CTAB, $R = 0.32$ solution, the behavior progresses from steady symmetric flow directly to unsteady flow. At a flow rate around $Q = 60$ mL h^{-1} ($\dot{\varepsilon} = 267\ s^{-1}$, $Wi = 2.1$) the flow becomes unsteady and a very faint birefringent band can be seen along the centerline of the outflow channel. Also, due to the unsteadiness, the intensity of the birefringent band grows and decays, sometimes to the point where the band can no longer be observed. Some sample birefringence images of the flow at $Q = 80$ mL h^{-1} ($Wi = 2.8$) are shown in Fig. 11 to exhibit the unsteady flow behavior: an image of the average birefringence, averaged over 100 images, and the deviation from this average at several instances in time.

The behavior of the 0.075 M CTAB, $R = 0.24$ solution is quite similar to that of the 0.050 M CTAB, $R = 0.32$ solution. At a flow rate of about $Q = 70$ mL h^{-1} ($\dot{\varepsilon} = 311\ s^{-1}$, $Wi = 2.5$) the flow transitions directly from steady symmetric to unsteady. For flow rates up to $Q = 20$ 0mL h^{-1} ($Wi \sim 7$) no vortices were observed with the 0.050 M CTAB solution or the 0.075 M CTAB, $R = 0.24$ solution.

Fig. 12 shows the region of the *Wi*–*Re* phase space where each of the phenomena appears. It can be seen that the shapes of the asymmetric flow region and the lip vortex region are qualitatively different.

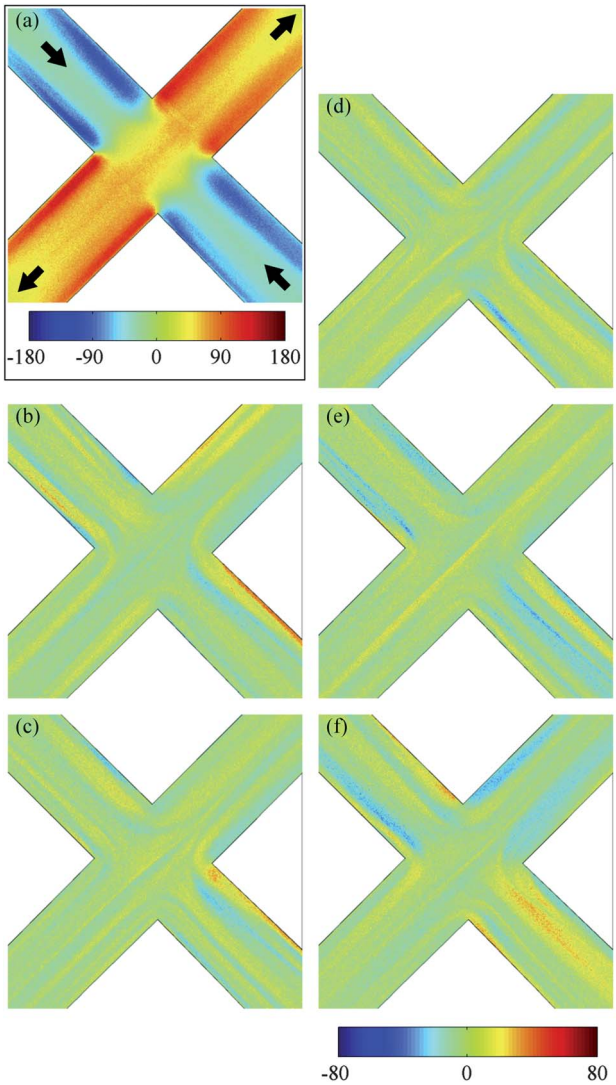

Fig. 11 False-color birefringence images for the 0.050 M CTAB, $R = 0.32$ solution at a flow rate of Q = 80 mL h^{-1}. Image (a) shows the birefringence intensity averaged over 100 images (the frame rate is 100 fps). Images (b)–(f) show the deviation in the birefringence intensity from the average at fixed times: (b) t = 0.0 s, (c) t = 0.1 s, (d) t = 0.2 s, (e) t = 0.3 s, and (f) t = 0.4 s. Fluid enters from top right and bottom left channels and exits from the top left and bottom right channels. In this case the intensity of the birefringence is roughly an order of magnitude less than for with the 0.100 M CTAB, $R = 0.32$ and 0.075 M CTAB, $R = 0.32$ solutions.

4 Discussion

The symmetry-breaking instability observed here has been seen previously in microfluidic cross-slot flows with a variety of viscoelastic fluids—experimentally, with polymer solutions[13] and micellar solutions,[14,15,36] and numerically with various viscoelastic constitutive models.[16,17,20] Additionally, flows of viscoelastic fluids in other microfluidic geometries have shown similar symmetry-breaking instabilities (for example ref. 19,21,25,26). In all these cases the Reynolds numbers were small and the instability was attributed to the elasticity of the material.

In some of the aforementioned works the flow would transition from steady symmetric to steady asymmetric to unsteady, in other cases the flow would progress directly from steady symmetric to unsteady. In our work, we have observed both these scenarios with different fluid classes. When compared to the numerical simulations that have been done, two observations can be made. Firstly, while the simulations where the steady asymmetric regime is observed by using different viscoelastic constitutive models (upper-convected Maxwell, FENE-P, FENE-CR), they are all in the regime where the solvent viscosity ratio, $\beta = \eta_s/(\eta_s + \eta_p)$ is small ($\beta < 0.2$), with η_s the solvent viscosity and η_p the polymer contribution to the viscosity. The work by Xi & Graham,[18] on the other hand, is in the dilute regime where $\beta \geq 0.95$, and they did not observe a steady asymmetric regime in the simulation. Instead, the flow transitioned directly to unstable flow at critical *Wi* values of around 60–80. Generally, it was observed that as β increased the critical

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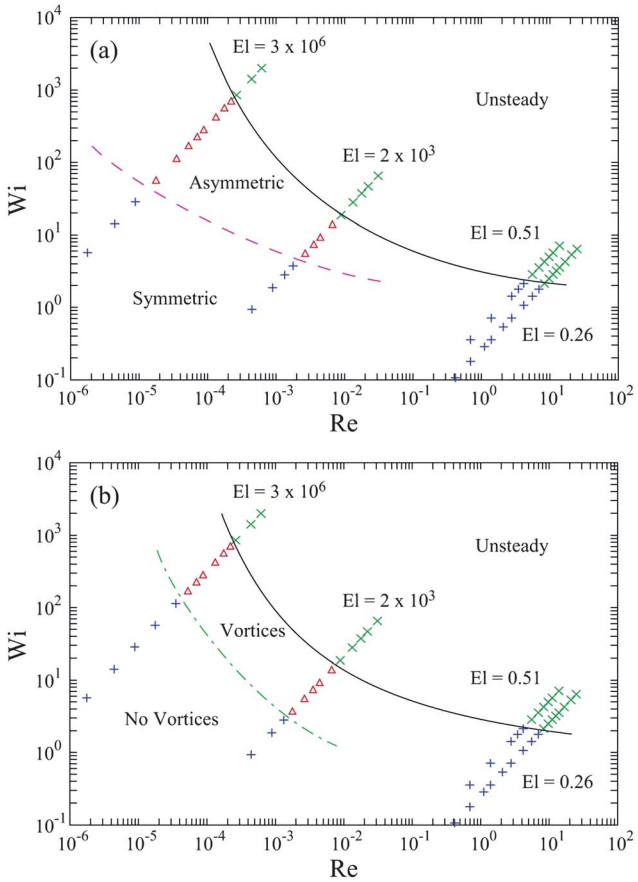

Fig. 12 (a) The *Wi–Re* phase space showing the steady symmetric flow region, the asymmetric flow region, and the unsteady flow region, pertaining to the symmetry-breaking instability. (b) The *Wi–Re* phase space showing the region in which the lip vortices are observed. In the unsteady region no lip vortices are observed.

Wi values also increased.[17,18] Secondly, there is a critical Reynolds number threshold ($Re \approx 2$–5) above which the steady asymmetric regime essentially disappeared.[16,21] Both these numerical simulation results are consistent with our experimental observations. For the weakly viscoelastic 0.050 M CTAB, $R = 0.32$ solution, the transition from steady symmetric flow to unsteady flow occurs at the Reynolds number $Re = 4.2$; and for the 0.075 M CTAB, $R = 0.24$ solution, the transition occurs at $Re = 9.7$, see Fig. 12(a). It has been shown[15] that Newtonian fluids in a microfluidic cross-slot device exhibit stable symmetric flow under Reynolds numbers of at least $Re = 15$. Thus the transition from the steady symmetric flow to unsteady flow of our weakly elastic solutions is likely an elastic effect. The combination of the elastic effect of the dilute solutions with increased inertial effects results in the disappearance (or bypass) of the steady asymmetric regime.

One discrepancy with all the previous work where a steady asymmetric regime was observed is that the critical *Wi* values at which the transition from steady symmetric flow to steady asymmetric flow occurred are higher here than previously reported. Generally, others found that this transition would occur when $Wi = 0.2$–2.5, while the critical *Wi* values of our studies range from 3.7–57. One possibility is due to the micellar solution used in this work. The rheology of micellar solutions is very complex, and the characterization of the rheology using a single relaxation time might not fully capture all the relevant factors for the instabilities. Different micellar solutions can have different behavior with respect to shear banding and micellar fracture and alignment. Hence, different micellar solutions (with the same λ) could have different critical *Wi* values. It has also been shown numerically that as the Reynolds number increases, the symmetry-breaking instability can be suppressed.[16] However, for our more elastic solutions (0.100 M CTAB and 0.075 M CTAB, $R = 0.32$ solutions), the transition from symmetric to asymmetric flow occurs at $Re \ll 0.01$ such that inertial effects should be negligible. Though, for our other two weakly elastic solutions, Reynolds numbers range over $Re = 0.1$–25 of flow rates examined (3–200 mL h^{-1}). Thus, for these solutions, inertial effects could be responsible for the lack of a steady asymmetric regime in those cases, see Fig. 12(a). In the experiments for both weakly and highly viscoelastic solutions, there are many important variables involved, such as the solvent viscosity ratio, the degree of shear banding, the degree of shear thinning, and strain thinning of the solution. In numerical simulations these quantities/parameters are usually kept as constants or not present in the problem. As a result, the full parameter phase space projections onto the *Wi–Re* subspace is not necessarily the same projections when comparing to *Wi–Re* phase space plots (*i.e.*, ones from numerical simulation and those from our experiments are different). Therefore a critical *Wi* trend in one projection might not look the same as that for a different projection. Nevertheless, our symmetry to asymmetry transition shows a similar trend to Rodd's observation *et al.*,[26] and our steady to unsteady transition has a similar trend to Haward *et al.*'s report.[36]

Another distinction between the previously reported work[13–15,36] and the work presented here corresponds to the roughness and surface properties of the microfluidic channel used. The influence of both the surface roughness and surface hydrophobicity on shear banding in rectilinear channel flow has been shown to affect both the wall shear stress and band width through numerical[37] and experimental investigation.[38] However, correlation between a quantitative measure of surface roughness or surface wetting is not available to make any connection between our channel or that of the previous work. Previously reported work used a combination of glass–PDMS[13,14] and glass–stainless steel[15,36] whereas we use only acrylic for our channel. Within and among these different channels, the surface roughness and also the surface wetting differ substantially. Thus, we do not expect our choice of material to have any significant contribution to the discrepancy with other work.

The formation of the lip vortices in our case is also an interesting occurrence in that lip vortices have generally not been observed in microfluidic cross-slot flow of viscoelastic fluids. In other microfluidic geometries such as contraction-expansion flows and 90° bends vortices have been observed and studied.[25–28] However, to our knowledge, no numerical simulations of microfluidic cross-slot devices have shown recirculating vortices, and only Haward *et al.*[15] have made any mention of lip vortices observed during experiments. Gulati *et al.*[28] found that vortices formed in a 90° bend at $Wi_{crit} = 0.84$ with DNA solutions and at $Wi_{crit} = 3.5$ with a polyethylene oxide (PEO) solution. In Haward *et al.*[15] the vortices are observed to form at

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$Wi_{crit} \approx 11$ which is higher than the critical Wi reported here ($Wi_{crit} = 3.7$ at $Re = 1.7 \times 10^{-3}$). Nevertheless, our lip vortices do show up at similar Wi and Re space (Fig. 12) in comparison to the data reported by Haward *et al.*,[15] with $Wi \sim 11$ at similar $Re \approx 9 \times 10^{-5}$. In the work by Gulati *et al.* it was also found that the vortices would only occur for shear thinning fluids. This is consistent with the result presented here; however, there is also a large difference in the Reynolds numbers between the experiments in which vortices did and did not occur. Hence it is not clear whether a lack of strong shear thinning is the cause for the lack of vortices seen with the weakly viscoelastic solutions.

Also of note is that the behavior of the symmetry-breaking instability and the vortex formation with respect to Wi and Re in the phase diagram are different. This is clearly seen in Fig. 12, where the phase diagrams are shown to be qualitatively different. For the 0.100 M CTAB, $R = 0.32$ solution the vortices appear at a critical Wi that is roughly 1.5 times larger than the critical Wi for asymmetry, whereas, for the 0.075 M CTAB, $R = 0.32$ solution the opposite is true, the critical Wi for symmetry-breaking is a factor of 1.5 larger than that for the vortex formation. Also, in the former case only two lip vortices form, whereas in the latter four vortices form. Thus, it seems that the vortex formation appears unrelated to the symmetry-breaking instability. However, we noticed that the critical Wi for this symmetry-breaking instability tends to decrease as the El decreases. This trend is opposite to what has been observed numerically by Afonso *et al.*[21] and the experimental work by Haward and McKinley,[36] but neither of them reported the appearance of any lip vortices. The experimental work of Rodd *et al.*[26] reported the symmetry-breaking trend that's consistent with ours. Furthermore, they also observed the appearance of lip vortices. This difference in trends suggests that the presence of lip vortices in our experiments may have an impact on the critical Wi at which the symmetry-breaking instability occurs. Additionally, this crossover of the critical Wi values between the 0.100 M and the 0.075 M CTAB, $R = 0.32$ solution suggests that our definition of Wi is likely not the proper choice for *both* phenomena. While the rate of extension (which is what is presently used to calculate Wi) seems a logical choice for the Weissenberg number which determines the transition to asymmetric flow, a more logical choice in terms of the vortex formation would be the wall shear rate. Nonetheless, if a nominal wall shear rate were to be used to calculate a "shear" Weissenberg number, it would at most differ by a constant factor from the Weissenberg numbers quoted here, and the phase diagrams would still remain qualitatively different. A more accurate method would be to determine the wall shear rates and stagnation point extension rates from measurements of the actual velocity in the channel for each of the different fluids, and then to use these values to calculate the critical Wi values.

On a final note, micellar breakage can produce various instabilities,[22,23] and thus could be a contributing factor to the phenomena described here. In strong shearing and extensional flows CTAB solutions have been observed to transition from fluid-like flow to solid-like fracture or tearing. This has been most commonly observed in flow around cylinders, where above a certain speed, a cutting or tearing behaviour is observed in the wake of cylinder.[22,39] We also observe this phenomenon with our own 0.100 M CTAB, $R = 0.32$ solution on the macroscale when an object is rapidly drawn through the solution. While nothing quite as drastic was observed in the microfluidic device, it has been shown in numerical simulations of flow in a cross-slot that micellar chain breakage events are localized near the stagnation point and also along the inlet channel wall just upstream of the corner.[40] The latter location is precisely where we see the formation of the vortices. While it is unlikely that these breakage events are the driving factor behind either of the phenomena, it is possible that they could play an important role in when the transitions occur.

5 Conclusions

In this work we presented results for the flow of a wormlike micellar solutions in a microfluidic cross-slot device. Four different concentrations of a CTAB–NaSal aqueous solution were tested; two that were strongly viscoelastic and shear-thinning, and two that were weakly viscoelastic and exhibited shear-thickening. With the strongly viscoelastic solutions an instability, where the flow field first becomes asymmetric and then becomes unsteady, was observed to occur. Moreover, a secondary flow, where lip vortices would form along the walls of the inlet channels, also occurred. However, between the two strongly viscoelastic fluids, the Weissenberg number at which each phenomena appeared was quite different, with the order of appearance of the symmetry-breaking instability and of the lip vortices reversing between the two fluids. This seems to imply that a single definition of Wi is insufficient to properly capture the behavior of these two phenomena, and that the phenomena are likely distinct instabilities. For the two weakly viscoelastic fluids, flow would transition directly form steady symmetric to unsteady asymmetric, likely due to inertial effects suppressing the steady symmetric flow regime. Again, the Wi values at which this transition occurred (in all four concentrations of the solution) varied significantly. This indicates that these instabilities are complex phenomena, whose behavior is likely influenced by a combination of factors such as Weissenberg number, inertial effects, and rheological behavior of the micellar solutions.

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